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# AI-Based Approaches for Predicting Electricity Demand at the Duques de Soria Campus

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***Abstract*— This paper presents a short-term electricity demand forecasting model for the Duques de Soria campus in Spain, using Artificial Intelligence, Random Forest (RF), Multi-Layer Perceptron (MLP), and Long Short-Term Memory (LSTM). Based on six years of hourly recordings from the smart meter installed on the campus. This study examines the impact of various features on prediction performance. After passing through model optimization. The RF model, with only historical load and hour features, achieved the best results with an MAPE of 9.51%, MAE of 5.95 %, while MLP showed strong robustness results with 9.66 %. LSTM required richer sequences and more training data to perform better. The study justifies that careful feature selection is just as crucial as algorithm choice. This work supports sustainable energy management on university campuses by enabling more accurate, data-driven forecasting.**

## I. INTRODUCTION

Over the past decades, the global electricity consumption has shown a significant and sustained increase, reaching 27,000TWh in 2023, nearly three times higher than it was in 1980[1]. All this growth is linked to the development of the world population, which rose from 4.4 billion to 8.05 billion during the same period [2]. At the same time becomes a vital component of socioeconomic development, and the demand is expected to rise continuously. That leads to the necessity of managing it, especially when it depends on fossil fuel-based power generation, which remains a major contributor to global $CO_2$ emissions [3]. In this context, electricity forecasting has an important role in achieving system stability, optimizing energy consumption, and reducing environmental impact[4]. Furthermore, to address the complexity of energy management, various solutions, including intelligent control, fault detection, and demand-side management strategies, have been developed and applied to mitigate this growing challenge and enhance building energy efficiency. Accurate prediction of building energy consumption is crucial for developing effective management strategies, achieving institutional sustainability targets, and optimizing the power system for reliable energy consumption prediction. This facilitates the system to adapt to avoiding energy waste and notifying the overcharging[5].The electricity waste is well illustrated by a study from Colmenar-Santos, which shows changing loads in time, avoiding energy waste and notifying the overcharging[5].The electricity waste is well illustrated by a study from Colmenar-Santos, which shows that more than 30% of building energy consumption can be saved if it can be reliably predicted [6]. So, it becomes crucial to anticipate future energy demands to support effective energy management strategies[7]. As mentioned earlier, robust forecasting can enhance energy efficiency and facilitate the integration of local renewable energy sources. University campus represents 50% of the electricity demand for the total buildings in Europe. Moreover, the energy behavior of such complex and activity-rich environments plays a vital role in shaping national consumption patterns [8]. Due to the growth of technology in recent years, intelligent artificial systems serve as a cornerstone of technological progress, driving advancements in virtually every sector. A lot of articles prove that the performance of forecasting using Machine Learning (ML)and Deep Learning (DL) is higher than using traditional models (statistical algorithms). The main objective of this study is to develop and evaluate an AI algorithm, such as machine learning and deep learning, to predict the next hours of electricity demand on a university campus in Soria, supporting energy management decision-making and optimizing the operation of the energy system. To achieve that goal, a comparison will be presented to compare the rentability of the chosen model using a real dataset, passing through Preprocessing of the data, Exploratory Data Analysis, Development and Comparison of Machine Learning Models to determine the Most Effective Approach for Smart Building for Accurate Forecasting. The study further investigates the impact of temporal, calendar, and weather features on prediction quality, thereby providing a comprehensive assessment of feature–model interactions

## II. DATA PREPARATION POSTPROCESSING

Collecting the electricity consumption data records from 2019 to 2025, with a resolution of one hour, from the university campus (Duque Soria), using smart metering to get the active energy consumed (kWh). It is essential to explore various types of input variables and identify the most significant ones. Moreover, weather-related data obtained via AEMET, a calendar that identifies exogenous features, is also available. Therefore, obtaining a daily weather dataset for Soria during the same period, which consists of (T(°C), precipitation (min), T(max), wind speed, solar peak hours, and



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relative humidity). Additionally, the university calendar categorizes it into working and non-working days. Without forgetting past series value(lags) variables that will be extracted from the historical data of consumption, moreover, extracting date from calendar such as Month, Year, Date, Time, Week, and Day, which makes it easy to analyze and to visualize. Referring to [9],[5] these explanatory variables are the most recommended for use in predictive models. As the technological universe progresses, electricity consumption behaviors change accordingly, depending on the needs. This is why the type of model changes over time, influenced by trends and seasonal variations. For that reason, chosen models need to be outlined to their influence and behaviors. In this case, the supervised machine learning approach was adopted by several previous studies, which demonstrate the performance of such models in forecasting electricity demand for a university campus, for example [9]. Analyse the performance of the model, according to the available data, showing that deep learning has a high-performance level. Instead, this article [10],pays attention to the importance of forecasting in the short term and its contribution to getting a straight management of electricity consumption in the same area. In addition is this study [5]they use the strategy of predicting 24-hour head considering the RF model as an efficiency model caused to its capacity to avoid overfitting by controlling the complexity of consumption data. on the other hand, the LSTM has a big change as a deep learning algorithm with its manipulation of memory [11].

*A. Random Forest (RF)*

Random forest regressor is a machine learning algorithm based on a combination of multiple decision trees to make a predictive model for regression tasks. The tree produces the output results from a randomly selected training subset, in addition to the selected features. The model gets a forest of trees and after that, voting for the best, averaging. It is more efficient for the Decision tree because it reduces the probability of overfitting by choosing the parameters controlling their depth and number of leaves by combining different trees. It is most used for its good ability to learn complicated behaviors and pattern identification [12].

*B. Multi-layer perceptron (MLP)*

Multi-layer perceptron (MLP) belongs to a classic family of Artificial Neural Networks (ANNs), presents an interconnected unit (neurons), and some neurons receive input from other neurons of the network to produce a single output. There are a wide variety of ANNs, the simplest and most used is the feedforward ANN. In this architecture, neurons are arranged in layers, one input one output and in the middle, there are one or more hidden layers with an information flow in one direction without feedback[13]. By applying weights from each input neuron process, adding them, and continuing to the results through a non-linear activation function that allows the network to learn complex relationships. These weights are adjusted during the training process to reduce the error of prediction and get the optimized model[3]. Furthermore, comparative analyses were performed, which showed that the MLP neural networks used in the regression problem provided better results than the MLP classifier models. [14]. So the main model is a multi-layer perceptron regressor.

*c. Long Short-Term Memory (LSTM)*

The long Short-term Memory (LSTM) model is introduced as an advanced application of neural networks (RNN), specially designed to capture the long-term interaction and handle sequences with variable length. Every time they need it, they just recall the previous data for this reason the LSTM is known as the robust model how collects and transmits trends from long sequences. Each neuron provides it, so in this case, the prediction result not only depends on the input and the weight but also the previous neuron inputs, acting like an internal memory. This enables it to model temporal relationships effectively. In conclusion, the internal architecture of the LSTM model comprises three main gates and one memory cell: the input gate, the forget gate, the output gate, and the cell state. These components work together to get the flow of information and preserve long-term trends across sequences [15].

*d.Evaluation metrics*

The splitting data is used to evaluate the performance by comparing the results of predictions given by the model after getting training, with the testing set that's reserved to get the evaluation of generalization capability and to avoid overfitting. Among these, different metrics are used according to the previous study [16], [17]. MAPE's (Mean Absolute Percentage Error ) presentation of absolute values makes it easy to interpret and evaluate the forecasting model's performance. It is selected for visually assessing the accuracy of the model's predictions, as it expresses error as a percentage. This means that MAPE can be used to compare the relative performance of different datasets, as it is represented by the equation (1).

$$\text{MAPE} = \frac{1}{n}\sum_{t=1}^{n}\left|\frac{x_t - y_t}{y_t}\right| \times 100\% \quad (1)$$

MAE ( Mean Absolute Error) serves to assess how the predicted values are too close to the real results. It is determined by averaging the absolute differences between the expected and the real values. The value near zero indicates better performance, while the higher the score, the worse the performance. And it is presented by equation (2).

$$\text{MAE} = \frac{1}{n}\sum_{t=1}^{n}|x_t - y_t| \quad (2)$$

Where $x_t$ is the actual value, $y_t$ is the predicted value, and n is the total number of estimations.

## III. Case Study Soria Campus

*A. Dataset description*

Based on the six-year data, which contains a total of 52677 of data entries, hourly records contain electricity consumption in (kWh), calendar variables (year, month, week, day, non-work ing day…). After that, the addition of the weather framework. To be more specific about the case study in Soria, some suppl ementary analysis is needed.



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*B. Visualization of the global time series*

The global electricity consumption is presented in Figure 1. Presents a random fluctuation, making it possible to identify visually general trends, seasonal variations, and consumption peaks linked to university activity. Further, some noise was presented.

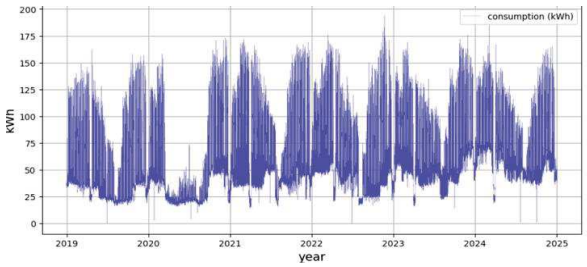



Figure1. Evolution of hourly electricity consumption from 2019 to 2024.

C. *Analyse statistique descriptive*

The curve generated the average daily load Figure 2 shows that the load consistently has two peaks, the first between 8h-13h and the second between 16h- 21h because of the extended study hours. This daily pattern is stable over the years, though from 2021 to 2024 the consumption is slightly higher, which can be due to an increase the campus activity and environmental factors. This analysis validates the importance of integrating temporal and historical context to get a robust forecasting framework.

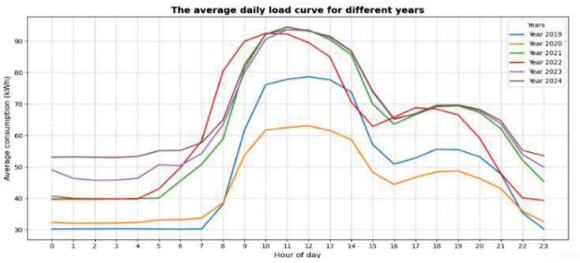



Figure 2. Average daily load curve through the years.

*C. Data Cleaning and Feature Engineering*

Data preparation is a key step in prediction to ensure it is ready for accurate analysis, as it may be incomplete or incoherent. Make sure that the missing raw data can enhance 0.5 % of the total value [18]. In this case, no missing values were noticed, which means a more forthright modeling process, eliminating the potential influence that could arise from handling incomplete data. Furthermore, the data needs to have the same proportion of repartition (min, hour…). This step preserves data integrity and enables the integration of multiple features to improve forecasting accuracy. Pre-processing approaches can be employed to implement some change in the data to get better performance for various models.

*1) Normalisation*

The purpose of data normalization is to bring the data back to narrow value ranges by adjusting it to a common scale. This is a necessary task to avoid premature function saturation of activations. It matters to reduce all the input datasets to the same order of magnitude, which improves the convergence

algorithm. Some models, especially deep learning models, need to be used after the normalization data [19]. Accordingly, the preprocessing includes various operations.

- Min-Max scaling normalization

In range normalization, the data is between 0 - 1, making the mean of the data 0 and the standard deviation 1. It is necessary to make the model fit better, allowing the model to learn more efficiently by reducing the influence of differing feature magnitudes. Represented by equation (3)

$$X=\frac{x-\min(max_{n-}\ min_{n}\ )}{max-min}+\ min_{n} \qquad (3)$$

With x being the original value, X is the normalized value, and (min, max) are the min and max values from the data.

- Standardisation

Also called mean normalization, using scale without losing information is the main idea of the standardisation method, to just put all the data between -1 and 1, which allows the models to work properly without any distribution[15]As it is represented in equation (4):

$$\text{ZScore} = \frac{x-x_{mean}}{x_{\sigma}} \qquad (4)$$

$$\text{Where } x_{\sigma} = \sqrt{\frac{1}{n-1}\sum_{i=1}^{n}(x_i - x_{mean})^2} \qquad (5)$$

$$\text{Such as } x_{mean} = \frac{1}{n}\sum_{i=1}^{n} x_i \qquad (6)$$

where x is the original value zscore is the normalized value(standardization), xmean is the mean value, $x_{\sigma}$ is the standard variation of x.

*2) Feature engineering*

- Weather data preparation

weather datasets need to be pre-treated and pre-selected (choose the best variable), since the number of input variables utilized in the modeling can affect the forecasting performance[14].

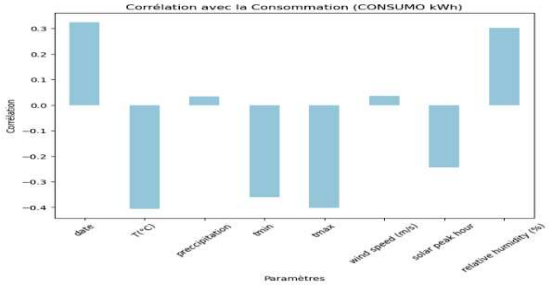



Figure 3. Correlation of electricity consumption and the climatological parameters.

A refugee to a correlation analysis of consumption data with the climatology parameters as it illustrated in figure 3 the temperature has shows the highly correlation compared to the others parameters as wind speed and the precipitation even it is negative correlation 0.4 but it remains significant so it potentially affect the results and it will be added to the dataset used in the modeling present in the next section.



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- Lag Feature Selection Strategy

Talking about lags, defining the past series values that correspond to the value of the data used before to predict these features helps improve the forecasting of the next hours[20].

Finding the optimal lag features starts by analyzing the autocorrelation (ACF) plots. The ACF presents a strong cyclic pattern every 24 hours, which means there's a daily seasonality.

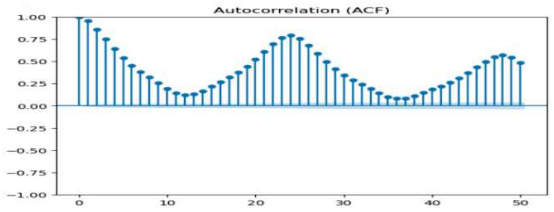



Figure 4. Autocorrelation (ACF).

Based on these insights, the lag values up to 24 were initially considered. To further refine this selection, we evaluated each forecasting model: Random Forest, MLP, and LSTM, using a range of lag configurations. Interestingly, the optimal number of lag features varied across the models. For example, the Random Forest model performed best with 20 lags, while the MLP and LSTM models yielded superior results with 24 lags, which aligns with the ACF analysis. This approach, combining statistical diagnostics with model validation, ensured that the selected lag structure was not only theoretically justified but also practically effective for each algorithm. Under these circumstances, recording hourly electricity usage provides detailed frequency, allowing for the identification of both short-term fluctuations and more extended usage patterns, which is particularly beneficial when building models to capture temporal dependencies.

*3) Time series forecasting strategies*

It is essential to formulate the techniques of the time series forecasting problem as a supervised learning problem. One common approach is the Sliding window technique, which converts sequential electricity consumption data into a tabular format by generating lagged features from previous consumption values. By moving a fixed-size window step by step through the dataset. Usually, this strategy is used to capture the temporal dependencies and recurring patterns in the dataset behavior. Tree strategies are frequently implemented in multistep forecasting. As shown, we can improve [11]. Direct strategy: models are trained to predict the future time step independently. It requires a complex architecture capable of multi-output prediction to get a better performance. Sequence-to-sequence: functions like a translation task, converting a sequence of past observations into a sequence of future values. It can capture the full temporal context of the input sequence and learn the long-range dependencies effectively. This strategy is more commonly used in long-term forecasting and scenarios where the relationship between input and output is sequential and nonlinear.

Rolling (recursive): knowing by an iterative method, predict one step at a time using the predicted output as input for the next prediction. This method is simple to implement and requires only a single model. This strategy is more susceptible to error accumulation through the sequence; nonetheless, it is particularly effective when strong intertemporal dependencies exist.

Although direct and sequence-to-sequence strategies offer alternative approaches for multi-step forecasting, the recursive method was selected due to the strong short-term dependencies in hourly electricity consumption. The direct strategy requires complex multi-output models and may fail to capture inter-hour correlations, while sequence-to-sequence models are typically more suitable for longer horizons or highly nonlinear patterns. In contrast, the recursive approach is straightforward to implement, requires only a single model, and effectively leverages the temporal correlations present in a 72-hour forecast, balancing simplicity and predictive performance [11].

## IV. RESULTS AND DISCUSSION

### *A. Performance comparison of Algorithms*

A comparative performance analysis of the three selected ML models applied to forecast the short-term electricity consumption across a multi-feature configuration. Input scenarios, to identify the optimal feature combination that yields the best performance, as represented in Table 1.all tested configurations, the scenario using only the historical load and the hour of the day as input features provided the most accurate results across the three models (RF, MLP, and LSTM), as illustrated in Figures 5 to 7

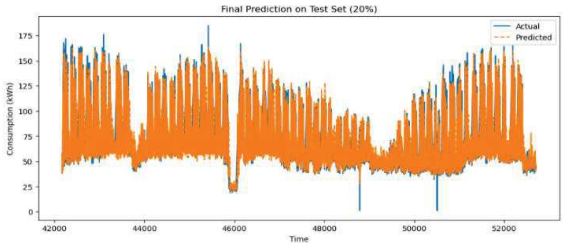



Figure 5. Predicted vs. Actual Energy Consumption (kWh) for MLP Model: Scenario-Based Load and Hourly Analysis.

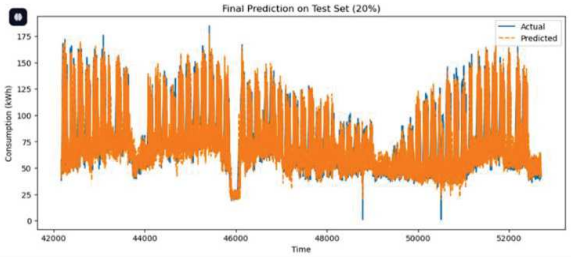
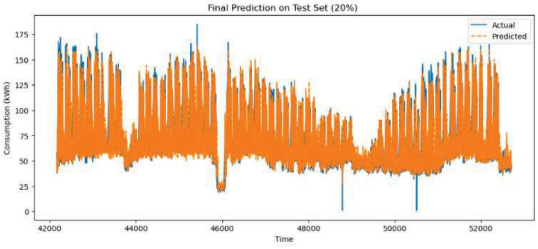



Figure 6. Predicted vs. Actual Energy Consumption (kWh) for RF Model: Scenario-Based Load and Hourly Analysis.






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Figure 7. Predicted vs. Actual Energy Consumption (kWh) for LSTM Model: Scenario-Based Load and Hourly Analysis

### *B. Error Analysis*

Table 1. MAE and MAPE in % of MLP, RF, and LSTM across different feature configurations

| Models / Scenario | MLPregressor | | RFRegressor | | LSTM | |
|---|---|---|---|---|---|---|
| | **MAE** | **MAPE%** | **MAE** | **MAPE %** | **MAE** | **MAPE %** |
| **Load** | 8.37 | 12.70 | 9.29 | 13.38 | 9.79 | 14.64 |
| **Load and hour** | 5.83 | 9.66 | 5.95 | 9.51 | 8.42 | 12.09 |
| **Load and day** | 8.95 | 13.88 | 9.34 | 13.46 | 10.61 | 15.03 |
| **Load and month** | 8.71 | 13.26 | 9.29 | 13.36 | 10.36 | 15.03 |
| **Load and year** | 8.82 | 13.65 | 9.35 | 13.50 | 11.09 | 15.63 |
| **Load and non working** | 8.71 | 14.02 | 9.40 | 14.18 | 9.71 | 14.39 |
| **Load and temperature** | 8.62 | 13.33 | 9.45 | 13.52 | 9.77 | 14.23 |
| **Load, year and month** | 9.10 | 14.17 | 9.34 | 13.47 | 10.20 | 14.53 |
| **Load, Month and Holiday** | 9.62 | 16.37 | 10.41 | 16.59 | 10.49 | 15.63 |
| **Load, month, non working and temp** | 9.97 | 17.18 | 10.58 | 17.14 | 11.48 | 17.94 |
| **Baselines** | MAPE= 18.90 | | | | | |

A deeper examination of the prediction error associated with each model across various feature configurations to identify the fundamental reasons, when and why certain models underperform. Certainly, there is a feature that outperforms, as it means there's no interaction that the model can find to get additional information, so that becomes a field of noise that can extend the margin of error. **Impact of lags**: The addition of historical value (electricity consumption) presents a significant implication for the model performance, as it is presented with lags only. MLP gives good results compared to the other models. due to its ability to learn nonlinear temporal patterns efficiently and its behavioral consistency.

**Impact of Temporal Features**: "Load + hour" yielded the lowest error for both RF and MLP, confirming that it is critical for short-term forecasting. Given the superiority of intraday variability (a seasonal trend as shown in Figure 2) for the other temporal features (month, year, etc.), the training model fails to capture operational schedules. Moreover, working and non-working days are affected, as the university campus calendar changes each year, and holidays do not occur on the same dates. This inconsistency weakens the ability of models to learn reliable seasonal or calendar-based patterns. Consequently, the model underperforms and exhibits a high error. **Impact of weather features**: Through the selected relevant meteorological factors, temperatures (°C) showed the highest correlation with electricity consumption. That can explain the satisfactory performance of the LSTM model. However, it still does not match the performance of other models. **Impact of feature combination:** As mentioned in the previous experiment, increasing the number of features does not necessarily improve the model's accuracy. In particular, scenarios that incorporate multiple features (month, year, holiday, temperature) increase model complexity without any improvement. This finding highlights the importance of feature selection in optimizing forecasting accuracy. To get an efficacy comparison, a baseline model affected using the last-day same-hour forecast to predict the hour by the consumption recorded at the same hour on the previous day referent to these results it is important to note that LSTM underperformed compared to the other models, explaining that by the small size of the data collected (dividing the six years into a sequence of time relatively limited for DL, which required a much larger training set. Also, using the recursive forecasting strategy adopted in this study is prone to error accumulation.

## V. Conclusion

Electricity demand is steadily increasing with rapid technological advances and urbanization. without accurate forecasting, this growth may lead to a shortage of electrical energy in the required quantity and lead to significant challenges for both producers and consumers, since electricity is a phenomenon that controls all aspects of life, accurate forecasting of electricity demand is a very important issue for policymakers as it allows better planning and management of energy resources ensuring energy security, efficiency, and sustainability. This investigation aims to develop an effective and realistic short-term electricity forecasting model. It has been prove that forecasting involves more than just advanced algorithms; it also encompasses the right features, interpretability, and domain adaptation. By combining rigorous experimentation with practical recommendations, we provide a framework for campus energy managers to optimize



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demand forecasting, reduce costs, improve sustainability, and support smarter grid integration. We utilized machine learning techniques (RF, MLP, LSTM) based on six years of hourly electricity consumption data from the Duque Soria University campus. We demonstrated the importance of data preprocessing, feature engineering, and systematic model evaluation in achieving accurate forecasts. Throughout the study, we established a methodological pipeline, from raw data cleaning to feature selection and model optimization, highlighting the impact of temporal, calendar, and weather-related variables on consumption behavior. Several feature scenarios were tested, with results indicating that RF maintains a strong baseline accuracy against noisy inputs, MLP provides robust generalization, while LSTM necessitates careful evaluation, as it did not perform well. Upon comparing the model results, we found that the best performance was by RF using hourly temporal features, achieving 9.51% MAPE and 9.95 MAE, significantly outpacing the other models across all datasets. Similar to the others, MLP achieved a low result by 9.98% MAPE and of MAE 5.83 using only hours and load, while LSTM requires larger datasets, giving 13.12% and 8.42 MAE in the same scenario, adding a high percentage of error. As energy systems grow increasingly complex, the key lessons from this study remain relevant: simplicity, transparency, and data-driven feature selection are the foundations of reliable predictions. Future advancements in AI should build upon these principles to create adaptable, efficient, and scalable solutions for the energy transition.

## REFERENCES


[1] Statista, "Global electricity consumption from 1980 to 2023,", id280704.

[2] Statista, "Total population worldwide from 1950 to 2100," Statistic ID 805044.

[3] F. Divina, M. G. Torres, F. A. G. Vela, and J. L. V. Noguera, "A comparative study of time series forecasting methods for short term electric energy consumption prediction in smart buildings," Energies (Basel), vol. 12, no. 10, 2019, doi: 10.3390/en12101934.

[4] S. Salehimehr, B. Taheri, and M. Sedighizadeh, "Short-term load forecasting in smart grids using artificial intelligence methods: A survey," The Journal of Engineering, vol. 2022, no. 12, pp. 1133–1142, Dec. 2022, doi: 10.1049/tje2.12183.

[5] D. Mariano-Hernández et al., "A data-driven forecasting strategy to predict continuous hourly energy demand in smart buildings," Applied Sciences (Switzerland), vol. 11, no. 17, Sep. 2021, doi: 10.3390/app11177886.

[6] X. Zhou, W. Lin, R. Kumar, P. Cui, and Z. Ma, "A data-driven strategy using long short-term memory models and reinforcement learning to predict building electricity consumption," *Appl Energy*, vol. 306, Jan. 2022, doi: 10.1016/j.apenergy.2021.118078.

[7] R. V. Klyuev *et al.*, "Methods of Forecasting Electric Energy Consumption: A Literature Review," Dec. 01, 2022, *MDPI*. doi: 10.3390/en15238919.

[8] P. Bastida-Molina, J. Torres-Navarro, A. Honrubia-Escribano, I. Gallego-Giner, and E. Gómez-Lázaro, "A detailed analysis of electricity consumption at the University of Castilla-La Mancha (Spain)," *Energy Build*, vol. 289, Jun. 2023, doi: 10.1016/j.enbuild.2023.113046.

[9] D. Mariano-Hernández *et al.*, "Comparative study of continuous hourly energy consumption forecasting strategies with small data sets to support demand management decisions in buildings," *Energy Sci Eng*, vol. 10, no. 12, pp. 4694–4707, Dec. 2022, doi: 10.1002/ese3.1298.

[10] W. Wu, Q. Deng, X. Shan, L. Miao, R. Wang, and Z. Ren, "Short-Term Forecasting of Daily Electricity of Different Campus Building Clusters Based on a Combined Forecasting Model," *Buildings*, vol. 13, no. 11, Nov. 2023, doi: 10.3390/buildings13112721.

[11] A. Mystakidis, P. Koukaras, N. Tsalikidis, D. Ioannidis, and C. Tjortjis, "Energy Forecasting: A Comprehensive Review of Techniques and Technologies," Apr. 01, 2024, *Multidisciplinary Digital Publishing Institute (MDPI)*. doi: 10.3390/en17071662.

[12] K. Elhabyb, A. Baina, M. Bellafkih, and A. F. Deifalla, "Machine Learning Algorithms for Predicting Energy Consumption in Educational Buildings," *Int J Energy Res*, vol. 2024, pp. 1–19, Jan. 2024, doi: 10.1155/2024/6812425.

[13] F. Pedregosa et al., "MLPRegressor – scikit-learn documentation (v1.4)," Scikit-learn.

[14] M. Sumorek and A. Idzkowski, "Time Series Forecasting for Energy Production in Stand-Alone and Tracking Photovoltaic Systems Based on Historical Measurement Data," *Energies (Basel)*, vol. 16, no. 17, Sep. 2023, doi: 10.3390/en16176367.

[15] S. Mahjoub, L. Chrifi-Alaoui, B. Marhic, and L. Delahoche, "Predicting Energy Consumption Using LSTM, Multi-Layer GRU and Drop-GRU Neural Networks," *Sensors*, vol. 22, no. 11, Jun. 2022, doi: 10.3390/s22114062.

[16] A. J. Niazai *et al.*, "Forecasting Monthly Electricity Consumption Using Machine Learning and Statistical-Based Models: A Case Study," 2022. [Online]. Available: https://www.researchgate.net/publication/377598332

[17] N. Tian, B. Shao, G. Bian, H. Zeng, X. Li, and W. Zhao, "Application of forecasting strategies and techniques to natural gas consumption: A comprehensive review and comparative study," Mar. 01, 2024, *Elsevier Ltd*. doi: 10.1016/j.engappai.2023.107644.

[18] D. Mariano-Hernández *et al.*, "Analysis of the Integration of Drift Detection Methods in Learning Algorithms for Electrical Consumption Forecasting in Smart Buildings," *Sustainability (Switzerland)*, vol. 14, no. 10, May 2022, doi: 10.3390/su14105857.

[19] Y.-S. Kim, M. K. Kim, N. Fu, J. Liu, J. Wang, and J. Srebric, "Investigating the Impact of Data Normalization Methods on Predicting Electricity Consumption in a Building Using different Artificial Neural Network Models.," *Sustain Cities Soc*, p. 105570, Jan. 2024, doi: 10.1016/j.scs.2024.105570.

[20] C. Kuster, Y. Rezgui, and M. Mourshed, "Electrical load forecasting models: A critical systematic review," Nov. 01, 2017, *Elsevier Ltd*. doi: 10.1016/j.scs.2017.08.009.




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